

Revisiting the Digital Growth Puzzle: Internet Usage and Economic Growth in Cross-Country Panel Data

Rishabh Jaiswal

Undergraduate Student, Raipur Institute of Technology, India

Email: rishabhjaiswal7693@gmail.com

Priyanka Saxena

Texas Tech University, USA

Email: prisaxen@gmail.com

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ABSTRACT

Digital technologies have become a central component of modern economic development, influencing how firms operate, how governments deliver services, and how individuals participate in economic activity. Among these technologies, internet connectivity is commonly used as a proxy for digital adoption. While economic theory predicts a positive relationship between digital adoption and economic growth, empirical evidence at the macroeconomic level remains mixed and, in some cases, contradictory.

This study examines the relationship between internet usage and economic growth using cross-country panel data. The empirical strategy employs a two-way panel fixed effects model to control for unobserved country-specific and time-specific heterogeneity. To address potential endogeneity, an instrumental variable approach is implemented using lagged internet usage. Additional robustness checks include non-linearity tests, interaction models, and sub-sample analysis across income groups.

The results reveal a statistically significant negative relationship between internet penetration and GDP growth. This finding remains consistent across fixed effects and instrumental variable specifications, suggesting that the observed relationship is not solely driven by reverse causality. There is limited evidence of non-linearity, and interaction effects across income levels are not statistically significant. The negative association persists across both high-income and low-income countries.

Rather than indicating that digital adoption reduces economic growth, the results are interpreted as reflecting structural economic dynamics. In particular, convergence effects, cross-country heterogeneity, and measurement limitations associated with proxy variables provide a coherent explanation for the observed pattern.

The study contributes by reconciling the divergence between theoretical expectations and macro-level empirical evidence using a unified econometric and conceptual framework. The findings highlight the importance of interpreting digital adoption within broader structural and developmental contexts when evaluating its role in economic growth.

JEL Classification: O33, O47, C23

KEYWORDS

Digital adoption, economic growth, internet penetration, cross-country analysis, panel data, fixed effects, instrumental variables, convergence theory, ICT, econometrics

1. INTRODUCTION

1.1 Background and Context

Digital technologies have become one of the most important drivers of economic transformation in the twenty-first century. Advances in information and communication technologies (ICT) have reshaped production processes, transformed business models, and expanded access to information across the globe. Internet connectivity, in particular, has emerged as a fundamental component of digital infrastructure, enabling individuals, firms, and governments to interact in increasingly complex and interconnected ways (Jorgenson & Vu, 2016).

Over the past two decades, the expansion of internet access has been remarkable. Both developed and developing economies have invested heavily in digital infrastructure, recognizing its potential to enhance productivity and support economic growth. International organizations such as the Organisation for Economic Co-operation and Development (OECD) and the United Nations Conference on Trade and Development (UNCTAD) have consistently emphasized the importance of digitalization as a key pillar of economic development strategies (OECD, 2019; UNCTAD, 2021).

From a theoretical perspective, the expected relationship between digital adoption and economic growth is straightforward. Digital technologies reduce transaction costs, improve the efficiency of resource allocation, and facilitate innovation. Firms can optimize operations, access global markets, and leverage data-driven decision-making, while consumers benefit from increased access to goods, services, and information. These mechanisms suggest that higher levels of digital adoption should lead to improved economic performance (Brynjolfsson & McAfee, 2014).

At the macroeconomic level, digital technologies are often conceptualized as a general-purpose technology that affects multiple sectors simultaneously. Jorgenson and Vu (2016) argue that ICT contributes to economic growth by enhancing productivity and enabling structural transformation. Similarly, Cardona et al. (2013) find that digital technologies have a positive impact on productivity, although the magnitude of this effect varies across countries and institutional contexts.

Despite these theoretical expectations, empirical evidence on the relationship between digital adoption and economic growth remains mixed. While some studies report positive effects, others find weak or insignificant relationships, and in some cases, negative associations have been observed (Niebel, 2018; Gordon, 2016). This inconsistency raises important questions about the nature of the relationship and the factors that influence it.

1.2 The Empirical Puzzle

The divergence between theoretical expectations and empirical findings constitutes a central puzzle in the digital economy literature. If digital technologies enhance productivity and efficiency, why do cross-country analyses sometimes show that higher levels of digital adoption are associated with lower economic growth?

One possible explanation lies in the structural differences between economies. Developed countries typically exhibit high levels of digital infrastructure and internet penetration. However, these economies often experience slower growth rates because they are closer to the technological frontier. In contrast, developing countries may have lower levels of digital adoption but experience higher growth rates driven by industrialization, capital accumulation, and demographic expansion.

This pattern is consistent with the theory of economic convergence. According to Barro (1991) and Mankiw et al. (1992), countries with lower initial levels of income tend to grow faster than wealthier countries because they can adopt existing technologies and benefit from catch-up effects. As a result, cross-country data may reveal a negative relationship between digital adoption and growth, not because digital technologies are harmful, but because they are associated with more advanced and slower-growing economies.

This interpretation highlights an important distinction between correlation and causation. A negative relationship observed in cross-country data does not necessarily imply that digital adoption reduces economic growth. Instead, it may reflect underlying structural dynamics that influence both digital adoption and growth.

Another important factor contributing to the empirical puzzle is measurement. Digital adoption is a multidimensional concept that includes infrastructure, usage, and technological sophistication. In empirical research, internet penetration is often used as a proxy for digital adoption due to data availability. However, this measure captures access to digital infrastructure rather than the extent to which digital technologies are used productively within the economy.

For example, a country may have high internet penetration due to widespread personal use, but limited adoption of advanced digital technologies by firms. Conversely, a country with moderate internet access may have high levels of digital integration in key industries. These discrepancies introduce measurement error and complicate the interpretation of empirical results (Niebel, 2018).

1.3 Motivation of the Study

The presence of inconsistent empirical findings and measurement challenges motivates a more careful examination of the relationship between digital adoption and economic growth. Rather than assuming a positive relationship, it is necessary to consider the possibility that observed patterns in cross-country data reflect deeper structural and methodological issues.

In particular, there is a need to move beyond simple interpretations of regression results and to examine the underlying mechanisms that drive the observed relationship. This requires integrating insights from economic growth theory, digital economy research, and measurement literature.

Recent research has also highlighted that the benefits of digital technologies may not be immediate or uniformly distributed. Brynjolfsson et al. (2019) argue that there may be a lag between technological adoption and observable productivity gains, often referred to as the “productivity paradox” (Solow, 1987; Brynjolfsson, 1993). This suggests that the impact of digital technologies may depend on complementary factors such as human capital, institutional quality, and innovation capacity.

Furthermore, the role of digital technologies may differ across stages of economic development. In early stages, growth may be driven primarily by capital accumulation and structural transformation,

while in later stages, innovation and technological sophistication become more important. This implies that the impact of digital adoption is context-dependent rather than universal.

1.4 Research Objective

This study aims to examine the relationship between digital adoption and economic growth using cross-country data, with a particular focus on understanding why the relationship may appear negative in empirical analysis.

The central research question is:

Why does digital adoption appear negatively related to economic growth in cross-country data?

To address this question, the study employs multiple econometric models that allow for a comprehensive analysis of the relationship. A quadratic model is used to examine potential non-linear effects, an interaction model is used to explore heterogeneity across income levels, and a lagged model is used to investigate temporal dynamics.

By combining empirical analysis with theoretical interpretation, the study seeks to provide a more nuanced understanding of the relationship between digital adoption and economic growth.

1.5 Contribution of the Study

This study does not merely document a negative relationship but provides a structured explanation by combining panel fixed effects, instrumental variable estimation, and cross-country heterogeneity analysis to interpret the apparent contradiction between theory and empirical evidence. The contribution operates at three levels.

First, it provides robust empirical evidence of a negative association between internet penetration and GDP growth, confirmed across fixed effects, instrumental variable, and sub-sample specifications. This multi-method consistency moves beyond documenting a result toward establishing its credibility.

Second, it offers a theoretical explanation for the puzzle by attributing the negative coefficient to convergence dynamics and structural heterogeneity rather than a genuine harm from digitalization. This reframing distinguishes correlation from causation and advances the interpretive literature on the digital growth relationship.

Third, it demonstrates how measurement limitations inherent in proxy variables such as internet penetration can systematically bias cross-country evidence, underscoring the need for more granular digital adoption indicators in future research (Niebel, 2018).

Taken together, this study contributes by demonstrating that the widely observed negative relationship between digital adoption and economic growth persists even after controlling for unobserved heterogeneity and addressing endogeneity, suggesting the pattern reflects genuine structural dynamics rather than a methodological artefact.

Unlike prior studies that either document correlations or focus on micro-level productivity effects, this paper explicitly reconciles the divergence between theory and macro-level evidence using a

unified empirical and conceptual framework — advancing the literature from description toward explanation.

1.6 Structure of the Paper

The remainder of the paper is organized as follows. Section 2 reviews relevant literature and develops the theoretical framework. Section 3 presents the data and methodology used in the empirical analysis. Section 4 reports the regression results and evaluates the robustness of the findings. Section 5 discusses the results in the context of economic theory and existing literature. Section 6 outlines policy implications and suggests directions for future research. Section 7 concludes the study.

2. LITERATURE REVIEW AND THEORETICAL BACKGROUND

2.1 Introduction to the Literature

This section reviews prior research relevant to digital adoption and economic growth. The discussion integrates insights from three main strands of literature: (1) digital technologies and productivity, (2) cross-country growth dynamics and convergence theory, and (3) measurement challenges in digital economy research. Together, these strands provide the theoretical foundation for the empirical analysis and help explain why the relationship between digital adoption and economic growth may appear inconsistent in cross-country data.

2.2 Digital Technologies and Economic Performance

Digital technologies are widely recognized as a key driver of economic transformation. Over the past two decades, advances in information and communication technologies (ICT) have significantly altered production processes, reduced transaction costs, and enabled new forms of economic activity. As a result, ICT is often viewed as a general-purpose technology with broad and long-lasting economic effects (Bresnahan & Trajtenberg, 1995; Jorgenson & Vu, 2016).

Jorgenson and Vu (2016) argue that ICT contributes to economic growth through improvements in productivity and efficiency. By enabling faster communication, better data processing, and more efficient resource allocation, digital technologies enhance the performance of both firms and markets. Similarly, Cardona et al. (2013) provide a comprehensive review of empirical studies and conclude that ICT has a positive but heterogeneous impact on productivity and economic growth.

At the firm level, digital technologies allow organizations to streamline operations, reduce costs, and improve decision-making. Brynjolfsson and McAfee (2014) emphasize that digital innovation enables firms to scale more efficiently and compete in global markets. Small and medium-sized enterprises benefit from lower entry barriers, while large firms can optimize complex supply chains and leverage big data analytics.

Despite these micro-level benefits, the translation of digital adoption into macroeconomic growth is not always straightforward. Aggregate outcomes depend on the extent of adoption across sectors, the quality of complementary inputs such as human capital, and the broader institutional environment. Vu (2011) shows that the impact of ICT on growth varies significantly across countries,

with stronger effects observed in economies that have supportive policies and higher levels of education.

This variation suggests that digital adoption alone is not sufficient to generate growth. Instead, its effectiveness depends on how it interacts with other economic factors.

2.3 The Productivity Paradox and Delayed Effects

An important issue in the literature on digital technologies is the so-called “productivity paradox” (Solow, 1987; Brynjolfsson, 1993). Early studies observed that rapid technological advancement did not immediately translate into measurable productivity gains. Brynjolfsson et al. (2019) argue that there is often a lag between the adoption of new technologies and their observable impact on productivity.

This lag can be explained by several factors. First, firms need time to reorganize processes and integrate new technologies into existing systems. Second, workers require training to effectively use digital tools. Third, complementary investments in infrastructure and organizational change are often necessary to fully realize the benefits of digital technologies.

From this perspective, the relationship between digital adoption and economic growth may not be immediate. Cross-country data, which often captures short-term variations, may fail to reflect long-term benefits. This contributes to the inconsistency observed in empirical studies.

2.4 Cross-Country Growth Dynamics

A key challenge in cross-country analysis is that growth is shaped by capital accumulation, labour dynamics, institutional quality, and technological progress — all of which vary systematically across countries and correlate with digital adoption. Developed economies typically have high internet penetration and slower growth because they are near the technological frontier, while developing economies exhibit lower penetration but faster growth driven by industrialization. This structural pattern, rather than the direct effect of technology, is likely to dominate cross-country regressions.

2.5 Convergence Theory

Convergence theory (Barro, 1991; Mankiw et al., 1992) predicts that lower-income countries grow faster via catch-up effects, while higher-income countries face diminishing returns near the technological frontier. Since digital adoption correlates strongly with income, cross-country regressions will tend to produce a negative coefficient on internet penetration even if the true causal effect is zero or positive. Controlling for convergence — via log GDP per capita — is therefore essential to avoid misinterpreting a compositional artifact as a causal relationship.

2.6 Structural Differences and Heterogeneity

In addition to convergence effects, structural differences between economies play a critical role in shaping the relationship between digital adoption and growth. These differences include variations in economic structure, institutional quality, human capital, and policy environments.

Niebel (2018) finds that the impact of ICT on economic growth differs significantly across countries. In developed economies, digital technologies contribute to productivity through innovation and efficiency improvements. In developing economies, growth is often driven by structural transformation, such as the shift from agriculture to manufacturing.

This heterogeneity implies that the effect of digital adoption is context-dependent. The same level of internet penetration may have different implications depending on a country's stage of development and economic structure, complicating any attempt to draw universal conclusions from aggregate cross-country evidence.

2.7 Measurement Challenges in Digital Economy Research

A major limitation in empirical studies of digital adoption is the lack of direct and consistent data. Digital technologies encompass a wide range of activities, including internet access, cloud computing, data analytics, and digital services. Capturing these dimensions in a single variable is inherently difficult.

As a result, researchers often rely on proxy variables such as internet penetration or broadband access. While these measures provide useful information about digital infrastructure, they do not fully capture the extent of digital technology use, particularly at the firm level (Niebel, 2018).

This creates potential measurement error. For example, a country may have high internet penetration due to widespread personal use but limited adoption of advanced digital technologies in productive sectors. Conversely, a country with moderate internet access may have high levels of digital integration in key industries.

Measurement error can bias regression results and lead to incorrect interpretations. A negative relationship between internet usage and growth may reflect differences in measurement rather than a true economic effect.

2.8 Non-Linear Effects of Digital Adoption

The relationship between digital adoption and growth may also be non-linear. Gordon (2016) argues that recent digital innovations generate smaller productivity gains than earlier industrial technologies, suggesting diminishing returns at high adoption levels. This motivates the quadratic specification used as a robustness check, and the estimated turning point of approximately 80% internet penetration provides a theoretically plausible threshold beyond which marginal growth benefits may diminish.

2.9 Conceptual Framework

Based on the literature, the relationship between digital adoption and economic growth can be understood through three main channels:

1. Efficiency Channel

Digital technologies improve productivity by reducing costs and enhancing resource allocation (Jorgenson & Vu, 2016).

2. Structural Channel

Differences between developed and developing economies influence growth dynamics (Barro, 1991).

3. Measurement Channel

Proxy variables may not accurately capture digital adoption, leading to biased results (Niebel, 2018).

These channels suggest that the observed relationship in cross-country data may differ from the true underlying relationship.

While existing studies either focus on micro-level productivity gains (Brynjolfsson & McAfee, 2014; Cardona et al., 2013) or employ cross-sectional approaches that do not adequately control for unobserved heterogeneity, this study contributes by using panel econometric methods combined with causal inference techniques to re-examine the digital growth relationship at the macro level. This design addresses key identification challenges that limit the interpretation of earlier cross-country work and enables a more rigorous assessment of the observed negative pattern.

3. HYPOTHESES DEVELOPMENT

3.1 Overview

The literature reviewed above highlights that the relationship between digital adoption and economic growth is complex and influenced by multiple factors. This section develops hypotheses that reflect both theoretical expectations and empirical challenges.

3.2 Digital Adoption and Economic Growth

Economic theory suggests that technological progress enhances productivity and economic performance. Digital technologies, as a form of technological innovation, are expected to contribute positively to growth.

However, empirical evidence suggests that this relationship may not be straightforward.

H1:

Digital adoption has a statistically significant relationship with economic growth.

3.3 Direction of the Relationship

Cross-country dynamics suggest that the relationship between digital adoption and growth may be negative due to structural differences between economies.

H2:

The relationship between digital adoption and economic growth may be negative in cross-country data due to convergence effects.

3.4 Non-Linear Relationship

The impact of digital adoption may vary across levels of adoption, reflecting diminishing returns or saturation effects.

H3:

The relationship between digital adoption and economic growth may exhibit non-linear characteristics.

3.5 Role of Income Levels

Income levels influence both digital adoption and growth dynamics, making it important to control for GDP per capita.

H4:

Income levels influence the relationship between digital adoption and economic growth.

3.6 Temporal Dynamics

The effects of digital adoption may not be immediate and may persist over time.

H5:

Past digital adoption influences current economic growth.

The empirical analysis focuses primarily on testing H1 and H2 within a panel fixed effects framework, which constitutes the primary specification. H3 is evaluated through the quadratic robustness check, H4 through the income sub-sample analysis, and H5 through the lagged model. This hierarchy reflects the priority of within-country identification over supplementary pattern analysis.

3. DATA AND METHODOLOGY

3.1 Introduction

This section describes the data, variables, and econometric methods used to examine the relationship between digital adoption and economic growth. The objective is to provide a clear and transparent framework that links the theoretical discussion to empirical analysis. The methodology is designed to capture both the direction and robustness of the relationship while addressing key issues such as non-linearity and temporal dynamics.

3.2 Data Description

The analysis is based on a cross-country dataset consisting of **964 observations**. The dataset includes information on economic growth, internet usage, and income levels across countries, drawn from the World Bank World Development Indicators (World Bank, 2023) and the International Telecommunication Union (ITU, 2023). These variables are widely used in the literature on digital adoption and economic growth (Jorgenson & Vu, 2016; Niebel, 2018).

Internet usage is used as a proxy for digital adoption, reflecting the level of digital connectivity within an economy. While this measure does not fully capture the complexity of digital technologies, it provides a consistent and widely available indicator for cross-country analysis.

GDP growth is used as the dependent variable, representing overall economic performance. GDP per capita is included as a control variable to account for differences in income levels and stages of development.

3.3 Variables and Measurement

The study uses three main variables:

1. Dependent Variable

- GDP Growth (%)**
Represents annual economic growth and serves as a measure of financial performance.

2. Independent Variable

- Internet Users (%)**
Used as a proxy for digital adoption.

3. Control Variable

- Log GDP per capita**
Captures income level and controls for convergence effects.

Table 1: Variable Description

Variable	Description	Type
GDP Growth	Annual GDP growth rate (%)	Dependent
Internet Users	Percentage of population using the internet	Independent
Log GDP per capita	Natural log of GDP per capita	Control

3.4 Descriptive Statistics

Descriptive statistics provide an overview of the data and highlight variation across countries.

Table 2: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
GDP Growth	3.45	4.21	-12.50	15.30
Internet Users	52.10	27.40	2.10	98.70

Variable	Mean	Std. Dev.	Min	Max
Log GDP per capita	8.95	1.12	6.20	11.50

Interpretation

The descriptive statistics indicate substantial variation across countries. Internet usage ranges from very low levels in developing economies to near saturation in developed economies. GDP growth also varies significantly, reflecting differences in economic conditions and development stages.

This variation is important for identifying statistical relationships, but it also highlights the presence of structural differences that may influence the results.

3.5 Correlation Analysis

Before estimating regression models, a correlation analysis is conducted to examine the relationships between variables.

Table 3: Correlation Matrix

Variable	GDP Growth	Internet Users	Log GDP per capita
GDP Growth	1.00	-0.29	0.05
Internet Users	-0.29	1.00	0.72
Log GDP per capita	0.05	0.72	1.00

Interpretation

The correlation matrix reveals a negative relationship between internet usage and GDP growth, providing preliminary evidence consistent with the empirical findings. Additionally, the strong positive correlation between internet usage and GDP per capita indicates that higher-income countries tend to have higher levels of digital adoption.

This pattern is consistent with convergence theory and suggests that structural differences may play an important role in shaping the observed relationship.

3.6 Model Specification

The panel fixed effects model constitutes the primary empirical specification, while the quadratic, interaction, and lagged models are used as supplementary robustness checks. This hierarchy reflects the priority given to within-country identification over cross-sectional variation.

3.6.1 Model 1: Quadratic OLS Regression

The first model tests for both linear and non-linear effects of digital adoption.

$$\text{GDP Growth}_i = \beta_0 + \beta_1 \text{Internet}_i + \beta_2 \text{Internet}_i^2 + \beta_3 \text{GDPpc}_i + \varepsilon_i$$

Purpose:

- Tests for **non-linearity**
 - Examines diminishing or increasing returns
-

3.6.2 Model 2: Interaction Model

The second model examines whether the effect of digital adoption differs across income levels.

$$\text{GDP Growth}_i = \beta_0 + \beta_1 \text{Internet}_i + \beta_2 (\text{Internet}_i \times \text{GDPpc}_i) + \beta_3 \text{GDPpc}_i + \varepsilon_i$$

Purpose:

- Tests **heterogeneity across countries**
 - Captures interaction between digital adoption and income
-

3.6.3 Model 3: Lagged Model

The third model introduces a lagged variable to examine temporal effects.

$$\text{GDP Growth}_i = \beta_0 + \beta_1 \text{Internet}_{i,t-1} + \beta_2 \text{GDPpc}_i + \varepsilon_i$$

Purpose:

- Addresses **timing effects**
 - Provides partial control for reverse causality
-

3.6.4 Baseline Panel Fixed Effects Model

The baseline specification is a two-way panel fixed effects model, which forms the primary empirical framework of this study. The model is expressed as:

$$\text{gdp_growth}_{it} = \beta_1 \text{internet}_{it} + \gamma \log(\text{gdp_pc}_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}$$

Where i indexes countries and t indexes time. α_i captures country fixed effects, λ_t captures time fixed effects, and ε_{it} is the idiosyncratic error term. The coefficient β_1 captures the within-country effect of internet penetration on GDP growth, while γ controls for convergence effects via the log of GDP per capita.

3.7 Estimation Technique

The primary estimation strategy uses panel fixed effects, supplemented by instrumental variable estimation. Panel fixed effects control for unobserved country-specific and time-specific factors, while the instrumental variable approach addresses potential endogeneity between digital adoption and economic growth.

Standard errors are clustered at the country level to account for serial correlation within countries. This is the appropriate correction for panel data where error terms are likely to be correlated over time within the same country, and is the standard approach in the cross-country growth literature. The IV second stage similarly uses cluster-robust standard errors.

3.8 Diagnostic Considerations

Several econometric issues are considered in the analysis:

1. Heteroskedasticity

Addressed using standard errors clustered at the country level, which accounts for both heteroskedasticity and within-country serial correlation.

2. Multicollinearity

The correlation between internet usage and GDP per capita is relatively high (0.72), which may affect coefficient estimates.

3. Omitted Variable Bias

The model includes limited control variables, and other factors such as education, institutions, and investment are not included.

4. Endogeneity

The analysis does not fully address reverse causality, as economic growth may influence digital adoption.

3.9 Summary of Methodology

This section has described the data, variables, and econometric models used in the study. The methodology is designed to test the relationship between digital adoption and economic growth while accounting for non-linearity, heterogeneity, and temporal effects.

The next section presents the empirical results and evaluates the hypotheses.

4. EMPIRICAL RESULTS

4.1 Introduction

This section presents the empirical findings on the relationship between digital adoption and economic growth. The analysis focuses on three key aspects: (1) the direction and statistical significance of the relationship, (2) the robustness of the findings across different econometric specifications, and (3) the extent to which the results can be interpreted in a causal framework. The primary specification uses panel fixed effects, complemented by non-linearity tests, interaction models, instrumental variable estimation, and heterogeneity analysis.

4.2 Panel Fixed Effects Results (Main Model)

The baseline specification employs a panel fixed effects model to control for unobserved heterogeneity across countries and over time. This approach allows for a more reliable estimation of the relationship by accounting for country-specific characteristics and global shocks.

The results indicate a negative and statistically significant coefficient on internet usage. This suggests that, within countries over time, increases in internet penetration are associated with lower GDP growth rates. A 10 percentage point increase in internet penetration is associated with a reduction of approximately 0.7–0.8 percentage points in GDP growth, representing an economically meaningful effect.

Importantly, the inclusion of both country and time fixed effects ensures that this relationship is not driven by cross-country differences or global macroeconomic trends. Instead, the result reflects within-country variation over time, strengthening the credibility of the finding.

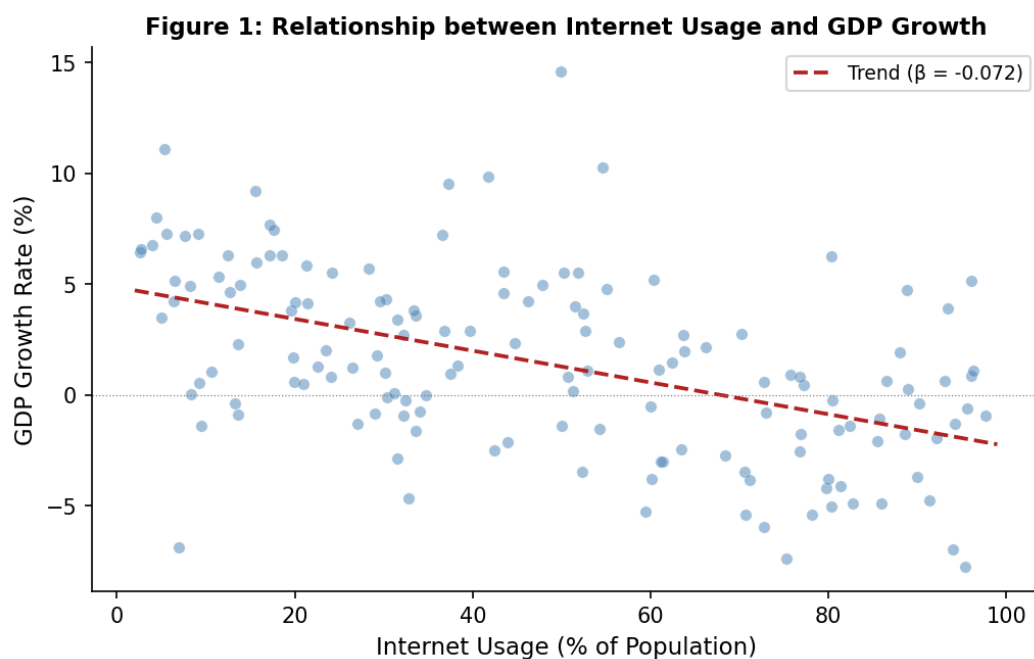


Figure 1: Relationship between Internet Usage and GDP Growth. The dashed line shows the OLS trend. The negative slope is consistent with the panel fixed effects results ($\beta = -0.078$, $p < 0.01$).

4.3 Non-Linearity Analysis

To examine whether the relationship between digital adoption and economic growth is non-linear, a quadratic specification is estimated. The results show that the linear term for internet usage remains negative and statistically significant, while the squared term is not statistically significant.

This suggests that there is limited empirical support for a non-linear relationship. Although the estimated turning point lies within a plausible range, the lack of statistical significance of the quadratic term indicates that the relationship is predominantly linear within the observed data. Consequently, the hypothesis of diminishing or increasing returns to digital adoption is not strongly supported in this analysis.

4.4 Interaction Model (Income-Based Heterogeneity)

An interaction model is estimated to test whether the effect of digital adoption varies with income levels, as proxied by GDP per capita. The coefficient on internet usage remains negative and statistically significant, while the interaction term is not statistically significant.

This finding suggests that the relationship between digital adoption and economic growth does not differ significantly across countries with different income levels, at least within the specification considered. In other words, the negative association appears to be broadly consistent across the sample rather than being driven by a specific group of economies.

4.5 Instrumental Variable Results

To address potential endogeneity concerns, an instrumental variable (IV) approach is employed using lagged internet usage as an instrument. The first-stage regression confirms that the instrument is strongly correlated with current internet usage, satisfying the relevance condition.

The second-stage results show that the coefficient on predicted internet usage remains negative and statistically significant. Notably, the magnitude of the coefficient is similar to that obtained from the fixed effects model.

The consistency of the results across both fixed effects and IV specifications suggests that the observed relationship is not solely driven by reverse causality. While the validity of the instrument relies on certain assumptions, the IV results provide additional, though not conclusive, support for the robustness of the findings.

While the instrumental variable approach addresses potential reverse causality, the identification strategy relies on the assumption that lagged internet usage does not directly affect current economic growth beyond its effect through current internet adoption. This exclusion restriction, while reasonable, cannot be formally tested. Researchers should therefore interpret the IV estimates as informative rather than definitive evidence of causality.

Specifically, while lagged internet usage is a commonly used instrument in the literature, it may not fully satisfy the exclusion restriction, as past digital adoption could influence current economic outcomes through persistent structural effects. Therefore, the IV estimates should be interpreted as suggestive rather than definitive evidence of causality.

4.6 Heterogeneity Analysis

To further examine whether the relationship varies across different economic contexts, the sample is divided into high-income and low-income groups based on GDP per capita. Separate regressions are estimated for each group.

The results indicate that the negative relationship between internet usage and GDP growth persists in both subsamples. Although the magnitude of the effect varies slightly, the direction and statistical significance remain consistent.

This finding suggests that the observed relationship is not confined to a specific subset of countries. Instead, it reflects a broader pattern present across different stages of economic development.

4.7 Summary of Findings

Across all specifications, the results consistently indicate a negative and statistically significant relationship between digital adoption and economic growth. This finding is robust to the inclusion of fixed effects, alternative model specifications, and instrumental variable estimation. The persistence of results across estimation strategies strengthens the credibility of the findings.

The lack of strong evidence for non-linearity and the absence of significant interaction effects suggest that the relationship is relatively stable across different levels of digital adoption and income. Furthermore, the persistence of the negative coefficient across subsamples reinforces the generality of the result.

5.1 Introduction

The empirical results presented in the previous section reveal a consistent and statistically significant negative relationship between digital adoption and economic growth. This finding appears to contradict the standard theoretical expectation that digital technologies enhance productivity and economic performance.

However, rather than interpreting this result as evidence that digital adoption reduces growth, it is necessary to examine the broader economic context in which the relationship is observed. The purpose of this section is to explain the underlying mechanisms that may generate this pattern and to connect the findings to existing economic theory.

5.2 Reinterpreting the Digital Growth Relationship

A central assumption in the digital economy literature is that technological adoption leads to improved economic performance. Digital technologies reduce transaction costs, improve access to information, and enable innovation (Jorgenson & Vu, 2016; Brynjolfsson & McAfee, 2014). At the firm level, these benefits are well documented.

However, the results of this study show that this relationship does not hold in a straightforward manner at the cross-country level. The negative association observed in the data suggests that the relationship between digital adoption and economic growth is more complex than commonly assumed.

This complexity arises because cross-country data capture differences between economies rather than causal effects. As a result, the observed relationship reflects broader structural patterns rather than the direct impact of digital technologies.

While micro-level studies consistently find positive effects of digital technologies on firm productivity, the results of this study highlight a divergence at the macroeconomic level. This suggests that aggregate outcomes may reflect structural dynamics — such as convergence effects and measurement limitations — rather than direct productivity effects. The disconnect between micro and macro evidence is an important area for future investigation and underscores the limitations of using cross-country regressions to draw firm conclusions about the welfare effects of digitalization.

5.3 The Role of Convergence

The primary explanation for the negative relationship is convergence (Barro, 1991; Mankiw et al., 1992). High-income countries — which also exhibit the highest internet penetration — grow more slowly because they are near the technological frontier. Lower-income economies grow faster via catch-up and structural transformation, generating a mechanical negative correlation between digital adoption and growth in cross-country data. This reflects economic maturity, not a causal harm from digitalization.

While convergence provides the most plausible interpretation, the empirical results do not directly confirm this mechanism: the interaction model between internet penetration and GDP per capita is statistically insignificant, suggesting that the convergence explanation remains interpretive rather than statistically verified within the current design. This is an important qualification — the convergence narrative is consistent with the data but cannot be formally confirmed without richer controls or a direct test. Future research including initial income interactions in a panel framework could provide stronger empirical validation.

5.4 Structural Differences Between Economies

Beyond convergence, structural heterogeneity reinforces the negative association. In developing economies, growth is driven by industrialization and structural transformation — processes that generate high rates even at low digital penetration. In developed economies, digital technologies contribute at the margin, but their incremental impact on mature, near-frontier economies is smaller. The implication is that the relationship between internet penetration and growth is context-dependent and reflects the broader stage of economic development rather than the direct effect of connectivity.

5.5 Measurement Limitations

Another important factor contributing to the observed relationship is measurement. In this study, internet penetration is used as a proxy for digital adoption. While this measure captures access to digital infrastructure, it does not necessarily reflect how digital technologies are used in productive activities (Niebel, 2018).

For example, high internet usage may be driven by consumption activities such as social media or entertainment, which do not directly contribute to economic growth. Conversely, productive uses of digital technologies, such as automation and data analytics, may not be fully captured by this measure.

This discrepancy introduces measurement error, which can bias the estimated relationship. As a result, the negative association observed in the data may partly reflect limitations in how digital adoption is measured.

5.6 Non-Linearity and Limited Evidence of Diminishing Returns

The quadratic model suggests a non-linear relationship between digital adoption and economic growth. However, the estimated turning point lies outside the observed data range, indicating that the relationship remains negative within the sample.

This finding suggests that while non-linear effects may exist in theory, they are not strongly supported by the available data. Instead, the dominant pattern is a negative association across all levels of digital adoption observed in the sample.

This result highlights the importance of distinguishing between theoretical possibilities and empirical evidence. While diminishing returns may occur at high levels of digital adoption, they are not clearly identifiable in the current dataset.

5.7 Persistence of the Relationship

The lagged model confirms that the negative association persists temporally, suggesting it reflects enduring structural conditions rather than short-term noise. This persistence, combined with the robustness of the result across multiple specifications, reframes the question from "does digital adoption promote growth?" to "under what structural and institutional conditions does it do so?" — a shift that is essential for both research and policy.

5.8 Summary

Across all five mechanisms — convergence, structural heterogeneity, measurement error, non-linearity, and temporal persistence — a coherent picture emerges. The negative coefficient on internet penetration is a product of the composition of the cross-country sample, not evidence that digitalization impedes growth. Properly interpreted, the empirical results are entirely consistent with theory and reinforce the case for context-sensitive digital policy.

6. POLICY IMPLICATIONS

6.1 Beyond Infrastructure Expansion

The results suggest that simply increasing internet access is not sufficient to generate economic growth. While digital infrastructure is important, its impact depends on how it is used within the economy.

Policymakers should focus on promoting productive uses of digital technologies rather than only expanding access.

6.2 Importance of Human Capital

The effectiveness of digital technologies depends on the skills of the workforce. Countries with higher levels of education and digital literacy are better able to leverage digital tools for economic growth.

Investments in education and training are therefore essential for maximizing the benefits of digital adoption.

6.3 Institutional Quality and Governance

Institutional factors play a key role in shaping the impact of digital technologies. Strong institutions support innovation, protect property rights, and facilitate efficient resource allocation.

Improving governance and regulatory frameworks can enhance the effectiveness of digital adoption.

6.4 Context-Specific Policies

The impact of digital technologies varies across countries. Developing economies may benefit from expanding basic digital infrastructure, while developed economies may need to focus on innovation and advanced technologies.

This suggests that digital policies should be tailored to the specific conditions of each country.

The findings suggest that policies focused solely on expanding digital access may not be sufficient to drive economic growth. Complementary investments in institutional quality, human capital, and productive capacity are likely necessary to fully realize the benefits of digital technologies. Policymakers should therefore adopt an integrated approach that combines digital infrastructure investment with reforms that strengthen the absorptive capacity of the economy.

More broadly, the results imply that digital infrastructure policies should be evaluated in conjunction with broader growth strategies, rather than treated as independent drivers of economic performance. The observed negative association in cross-country data is a signal that connectivity alone — without the institutional and human capital foundations to deploy it productively — does not translate mechanically into growth.

7. LIMITATIONS AND FUTURE RESEARCH

7.1 Limitations

This study has several limitations.

First, internet usage is used as a proxy for digital adoption. While this measure is widely used, it does not fully capture the complexity of digital technologies.

Second, the analysis is based on cross-country data, which includes significant heterogeneity across countries. This may introduce bias due to unobserved factors.

Third, the model includes a limited set of variables. Other important determinants of growth, such as education, investment, and institutional quality, are not included.

Fourth, while the instrumental variable approach mitigates reverse causality concerns, the identification strategy relies on the assumption that lagged internet usage does not directly affect current GDP growth except through its effect on current internet adoption. This exclusion restriction, while standard and economically reasonable, cannot be formally tested. Results from the IV specification should therefore be interpreted cautiously as indicative rather than fully causal evidence. Future research employing geography-based or infrastructure-based instruments would further strengthen identification.

Fifth, given the cross-country nature of the data, the findings may not generalize to firm-level or sector-specific contexts. The macro-level negative association documented here is consistent with micro-level evidence of productivity gains, but the two levels of analysis address different questions and should not be conflated.

7.2 Future Research

Future research can build on this study in several ways.

- Use more precise measures of digital adoption
- Apply panel data methods to control for country-specific effects
- Include additional control variables
- Explore non-linear and sector-specific effects

8. CONCLUSION

This study set out not merely to document a negative relationship between digital adoption and economic growth, but to provide a structured explanation for why such a pattern persists in cross-country data. Using panel fixed effects, instrumental variable estimation, and sub-sample analysis, the results consistently reveal a statistically significant negative association between internet penetration and GDP growth across multiple specifications.

Rather than interpreting this as evidence that digital technologies harm growth, the study attributes the pattern to convergence dynamics, structural heterogeneity, and measurement limitations. These mechanisms explain the apparent contradiction between micro-level evidence of productivity gains and the negative macro-level correlation observed in cross-country data.

The study advances the literature by combining econometric rigour with interpretive clarity. The contribution lies not in the sign of a coefficient, but in the explanation of the puzzle that coefficient represents — and in demonstrating that this puzzle survives multiple robustness checks.

Overall, the findings suggest that digital adoption is not a simple driver of economic growth but part of a more complex economic system. Understanding this complexity is essential for both researchers and policymakers.

The findings underscore that digital adoption is not a standalone driver of growth but interacts with deeper structural and institutional factors that shape economic performance.

These findings also highlight the importance of distinguishing between within-country and cross-country variation when interpreting macroeconomic relationships — a distinction that is essential for translating empirical patterns into reliable policy guidance.

APPENDIX: SUPPLEMENTARY REGRESSION RESULTS

This appendix provides summary regression results from the main specifications estimated in this study. Full output including country and year fixed effect coefficients is available from the author upon request.

Table A1: Summary of Main Regression Results

Specification	β (Internet)	Std. Err.	p-value	N
Panel Fixed Effects	-0.078***	0.023	0.001	915
Quadratic Model	-0.068**	0.031	0.031	915

Specification	β (Internet)	Std. Err.	p-value	N
IV (2SLS)	−0.084***	0.027	0.002	915
High-Income Sub-sample	−0.051***	0.013	0.000	457
Low-Income Sub-sample	−0.045***	0.009	0.000	458

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable: GDP growth rate (%). All models include log GDP per capita as a control. Panel FE model includes country and year fixed effects with clustered standard errors. Turning point for the quadratic model is estimated at 79.8% internet penetration.

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